

Computing Platform for Autonomous Driving (CP4AD)

1. Course Introduction

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Dept. Automobile and IT Convergence



Introduction to myself

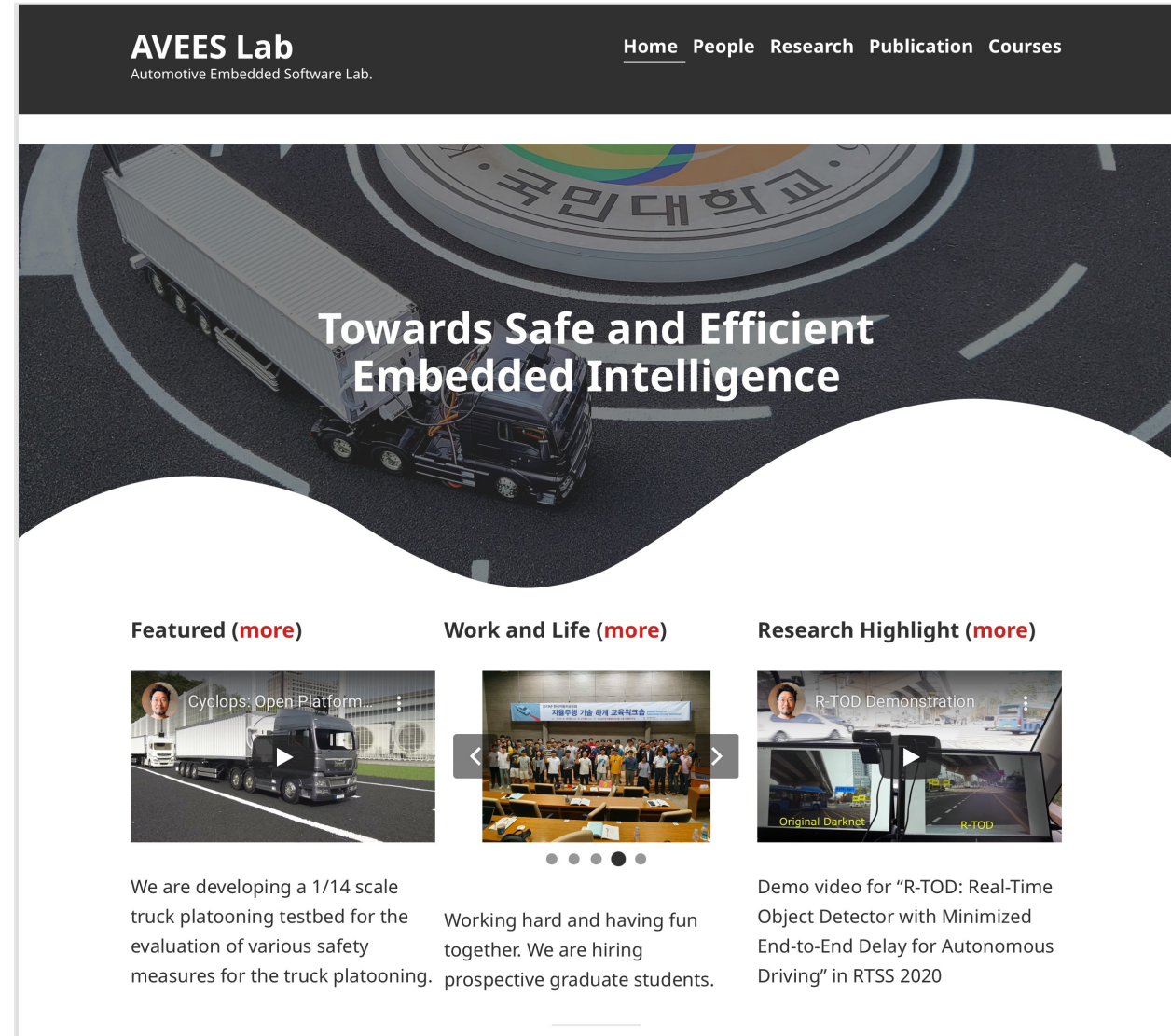


**Jong-Chan
Kim**

- PhD in computer science
- Research topics
 - Operating system
 - Real-time system
 - Computing platforms for autonomous driving
- Short-term career objective
 - **Make automotive control systems safer and more efficient with computer systems techniques**
- Long-term career objective
 - **Make automotive control systems safer and more efficient by training students into great engineers**
- Contact
 - Email: jongchank@kookmin.ac.kr

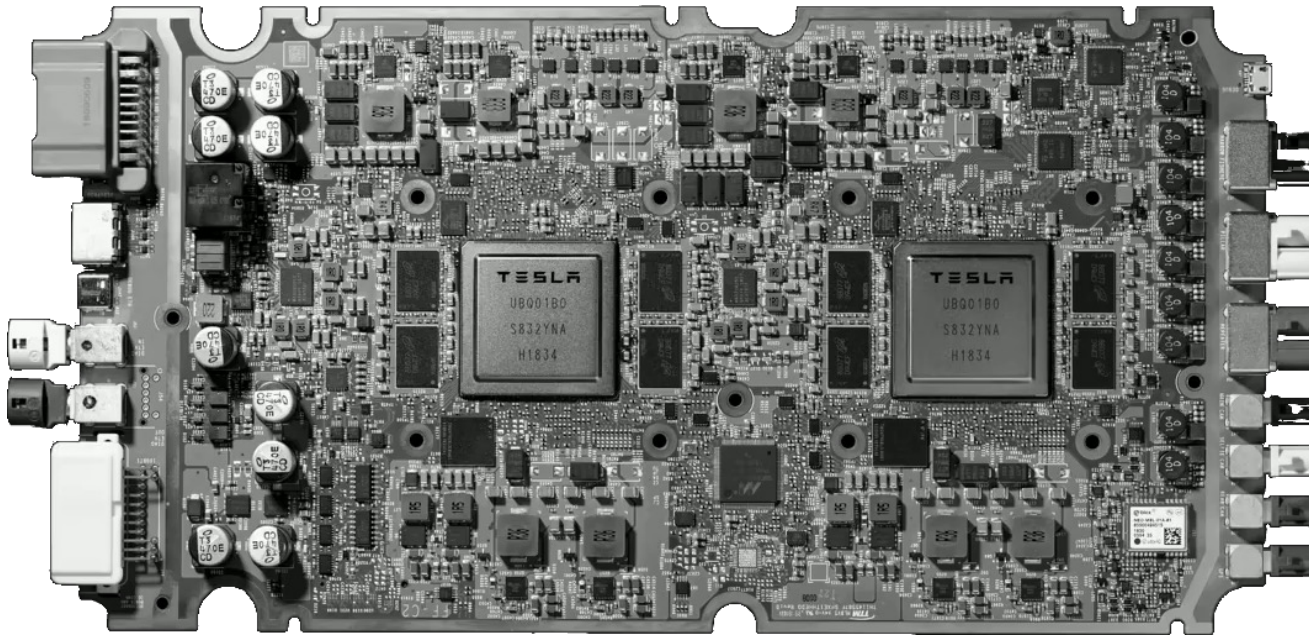
AVEES Lab. (<http://avees.kookmin.ac.kr>)

- Truck Platooning
 - [ICCAS '20][ICRA '22][DATE '23]
- Real-Time DNN Inference
 - [RTSS '20][RTSS '22] [ICCAS '23]
 - [ML4AD '23]
- AUTOSAR System Optimization
 - [ICCAS '18][Appl. Sci. '20]
- Energy-aware System Adaptation
 - [ISORC '21][TECS '23]
- Autonomous Driving
 - [TECS '21] [Appl. Sci. '22]

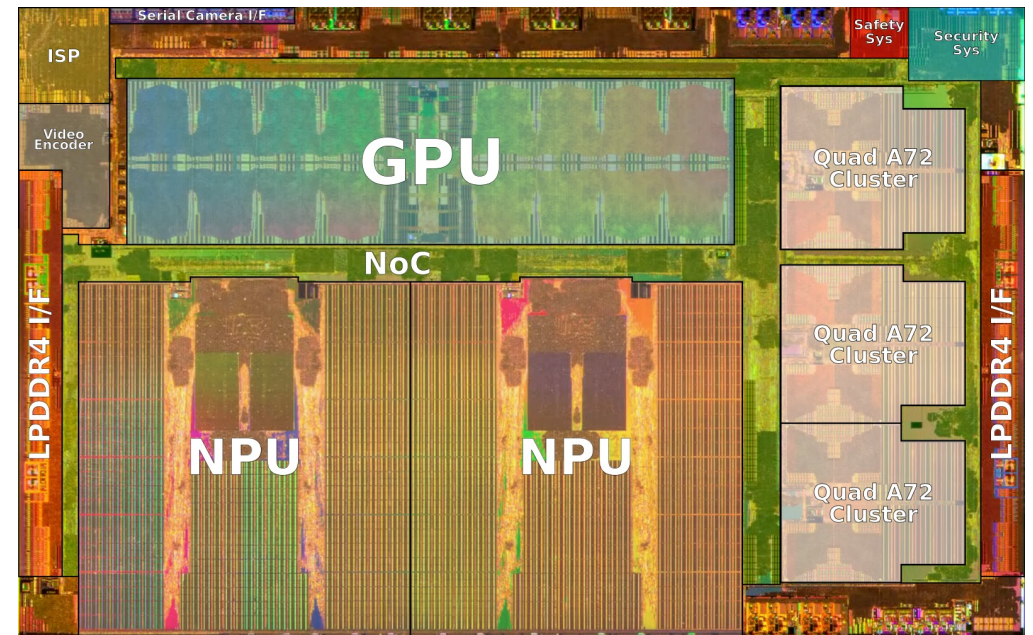


What this course is about

- To understand computing platforms (HW/SW) for autonomous driving
- How?
 - Learning by hands-on experiences



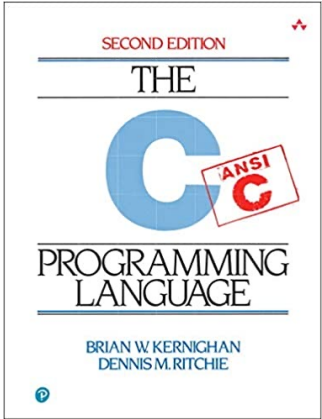
Tesla FSD (Full Self Driving) computer



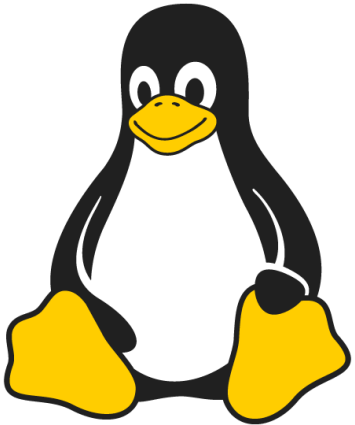
Its die photo

Prerequisites

- Understanding of the C language
- Experience in VIM



- Experience in Linux

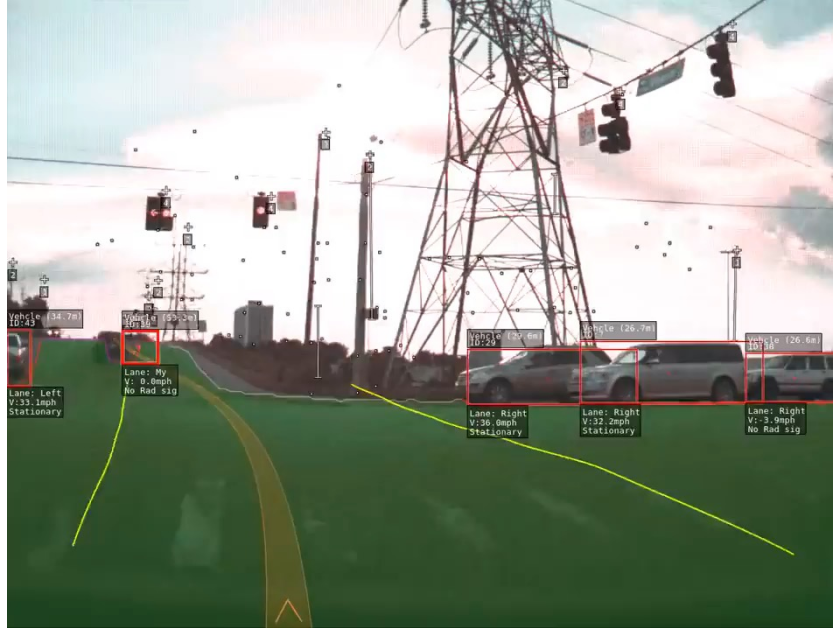
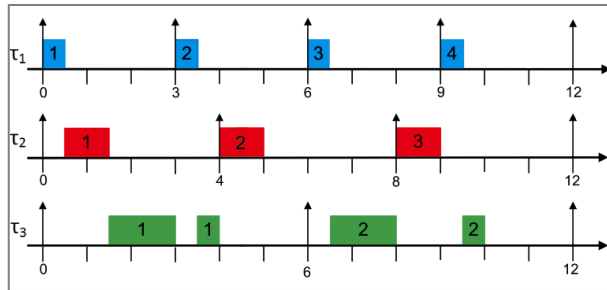
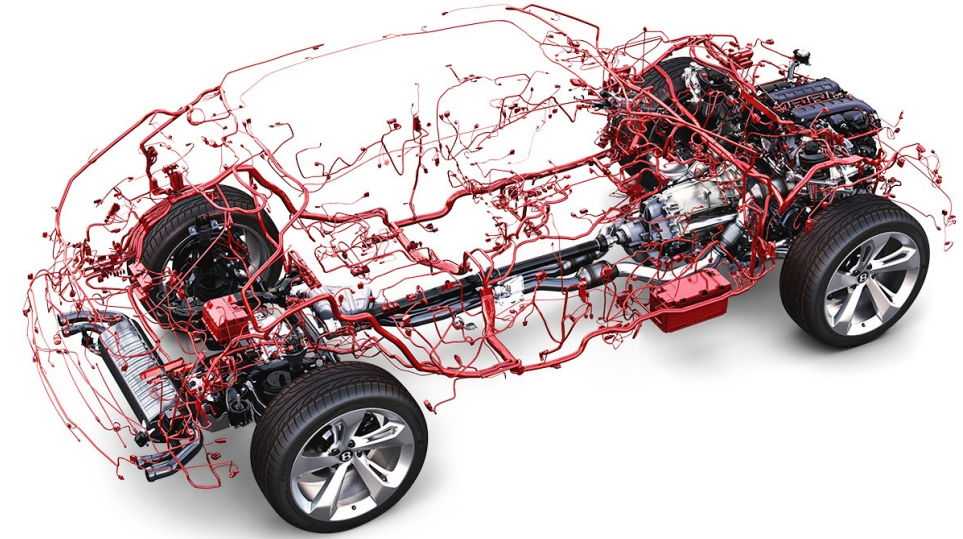


- Experience in GCC

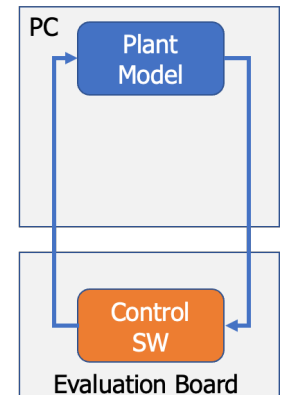
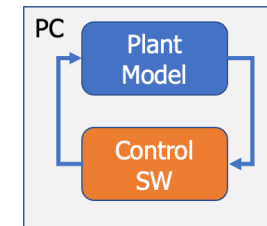
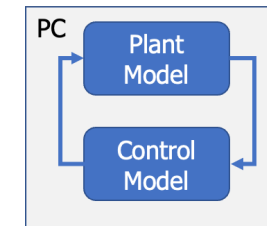


Hands-On Topics

- ① AUTOSAR OS (OSEK RTOS)
- ② On-Device AI
- ③ Controller Area Network (CAN)
- ④ Model-Based Development (MBD)



MIL (Model-In-the-Loop) SIL (SW-In-the-Loop) PIL (Processor-In-the-Loop)

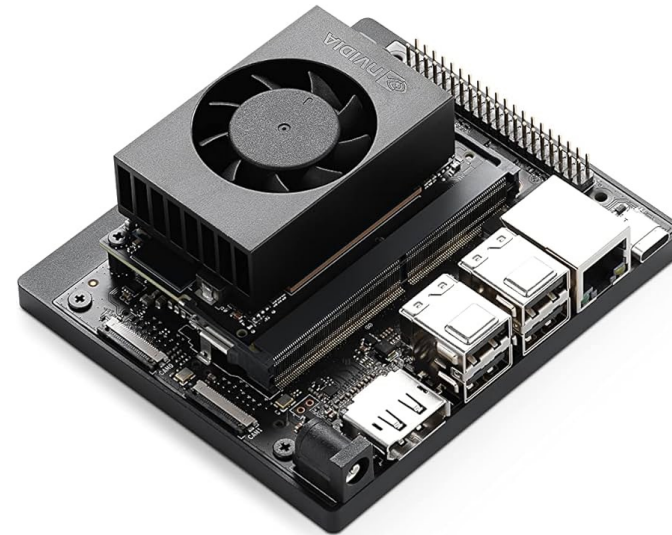
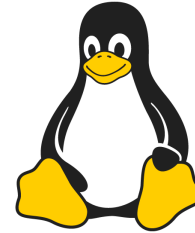
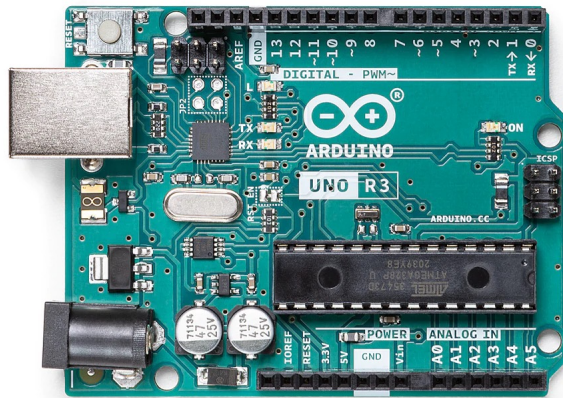


Hands-On Platforms

- Hands-on platforms
 - Arduino
 - AUTOSAR-compliant OSEK RTOS
 - Nvidia Jetson Orin Nano
 - Ubuntu Linux

AUTOSAR

ERIKA³



Logistics

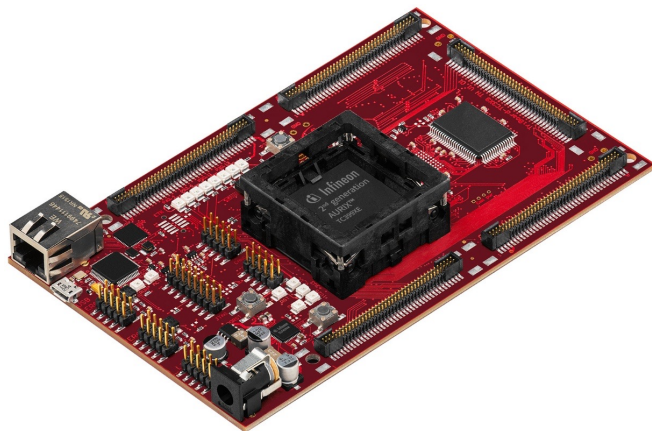
- Lecture
 - Thu (14:00~17:45)
- Grading
 - Mid-term exam (20%)
 - Final exam (20%)
 - Attendance (20%)
 - Participation (20%)
 - Project (20%)
- Office Hour
 - Mon (15:00 – 16:00)
- TAs
 - 김재혁 <pocarisweat257@kookmin.ac.kr>
 - 김다현 <kd4hyun95@kookmin.ac.kr>
 - 송원석 <sws9912@kookmin.ac.kr>
 - 김은우 <ox0203@kookmin.ac.kr>

Computing Platforms for Automotive Systems

Control

- (Multicore) CPU
- Controller Area Network (CAN)

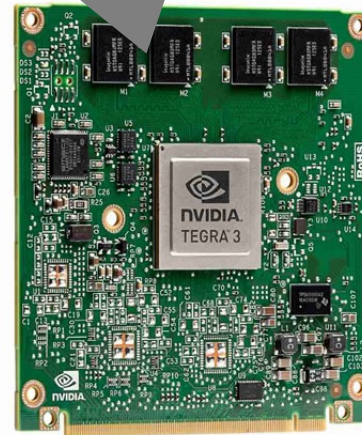
- Functional Safety (ISO 26262)
- Multicore Validation and Optimization
- RTOS (e.g., OSEK or AUTOSAR OS)



Infotainment

- + GPU (Graphics Processing Unit)
- Wireless (LTE, 5G, Bluetooth, ...)

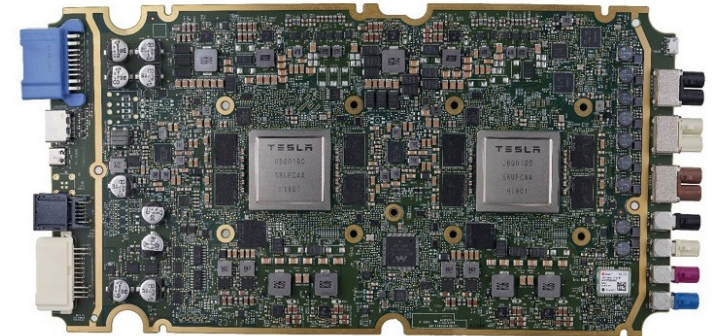
- + Cyber Security (ISO/SAE 21434)
- Efficient Graphics Processing
- Linux (e.g., Android Automotive OS)



Intelligence

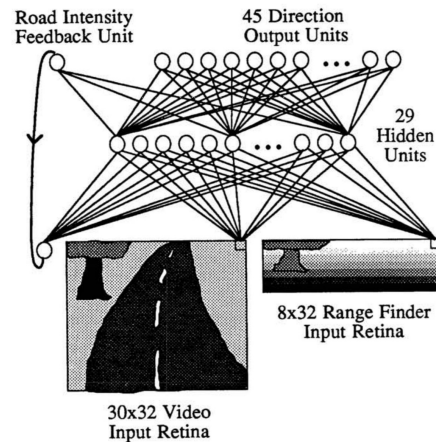
- + NPU (Neural Processing Unit)
- Ethernet

- + Intended Safety (ISO/PAS 21448)
- Efficient Deep Neural Network Inference
- POSIX OS (e.g., Linux and QNX)



Evolution of Automotive Intelligence Platforms

CMU ALVINN
(1988)

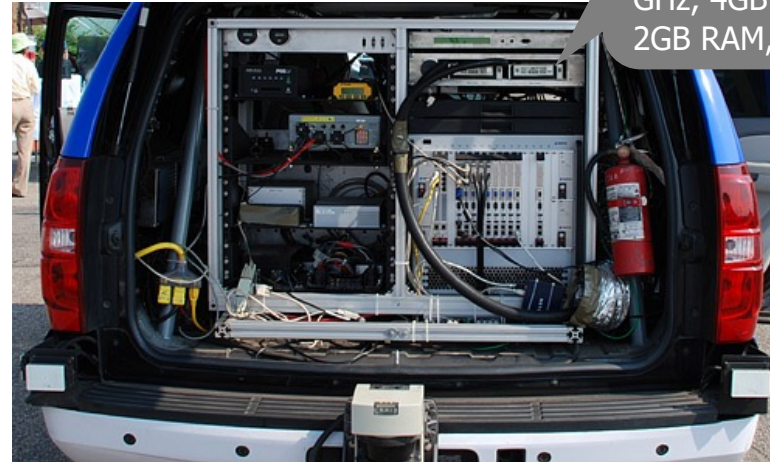


- 2 layer fully-connected network
- 4,000 parameters
- 5 minutes of training data

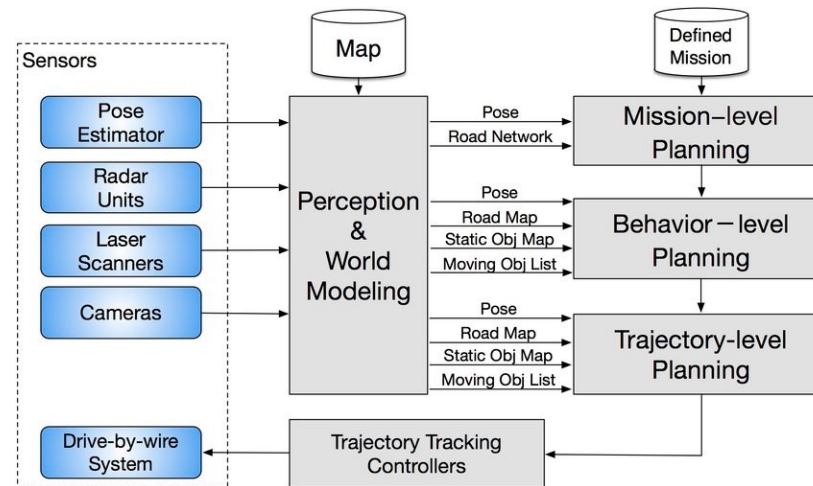
[Pomerleau, NIPS 1988]

3 Sun Workstations

CMU Boss
(2008)



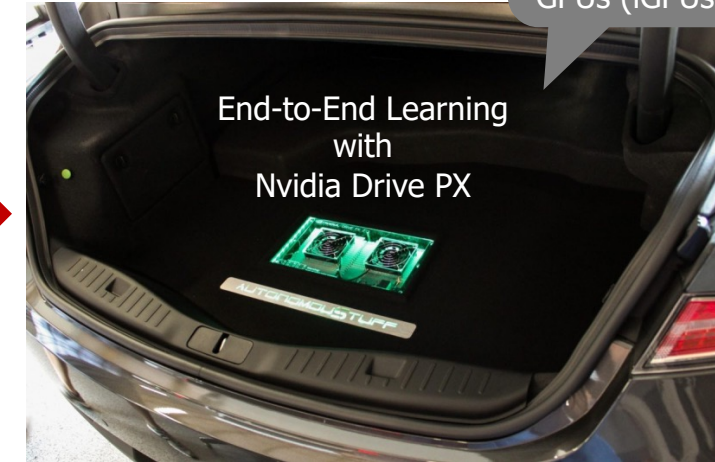
10 CompactPCI boards
(Core2Duo 2.16-GHz, 4GB ROM, 2GB RAM, gigE)



- No Neural Network. Before Deep Learning

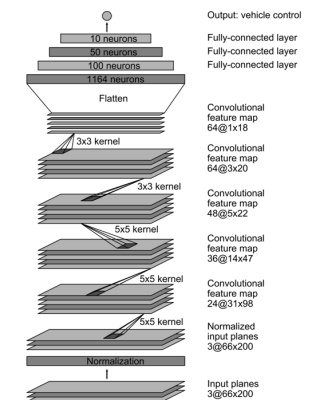
[Urmson et al., Journal of Field Robotics 2008][Gu, 2017]

Nvidia DAVE-2
(2016)



2x Tegra X1 SoC with integrated GPUs (iGPUs)

End-to-End Learning with Nvidia Drive PX



- 8-layer CONV and fully-connected network
- 250,000 parameters
- 3,000 hours of training data

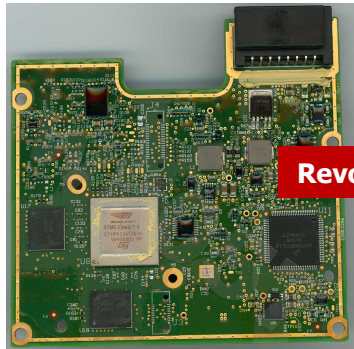
[Bojarski et al., ArXiv 2016]

Tesla's Intelligence Platforms

- Fast evolution and revolution for the recent years

- Mobileye EyeQ3 Vision Processor
- Freescale Safety Processor

HW 1.0



(2015)

Revolution

- 1X Parker SoC
- 1X Pascal GPU
- Infineon Safety Processor

HW 2.0

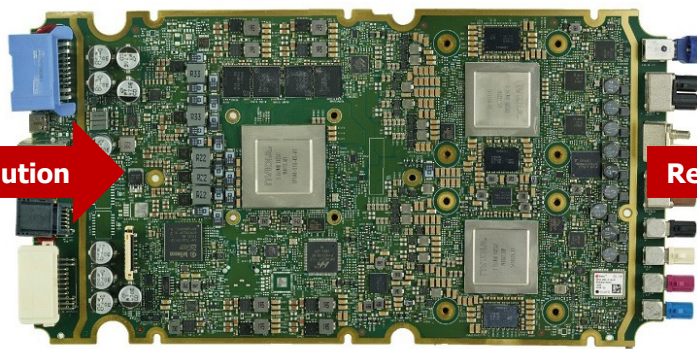


(2016)

Evolution

- 2X Parker SoC
- 1X Pascal GPU
- Infineon Safety Processor

HW 2.5

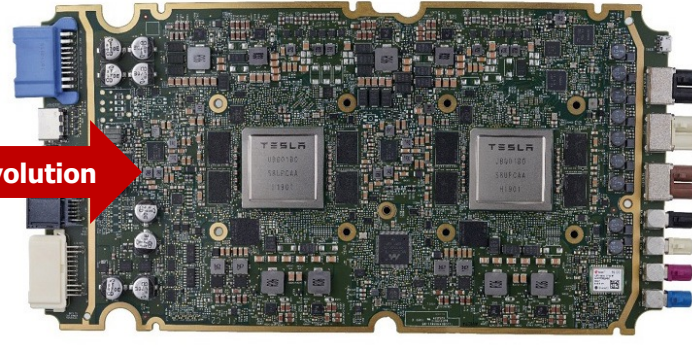


(2017)

Revolution

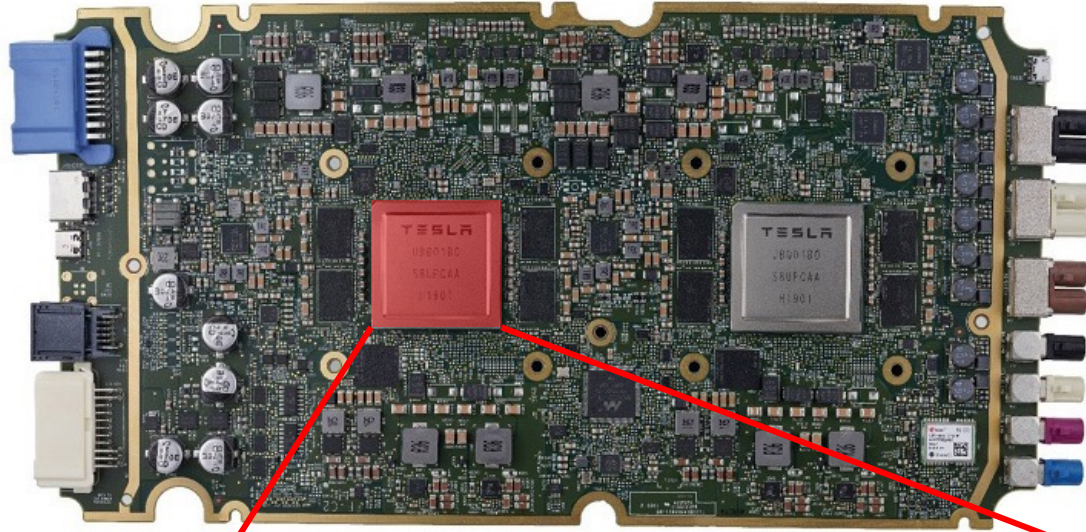
- 2X FSD (Full Self Driving) Chip
- No More Safety Processor

HW 3.0

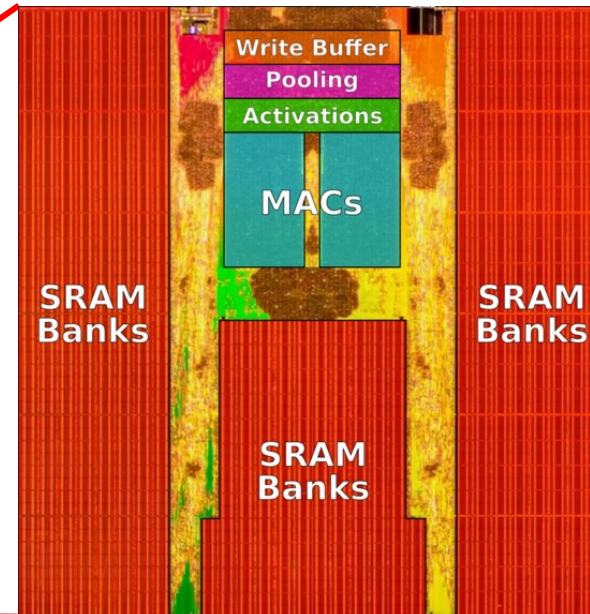
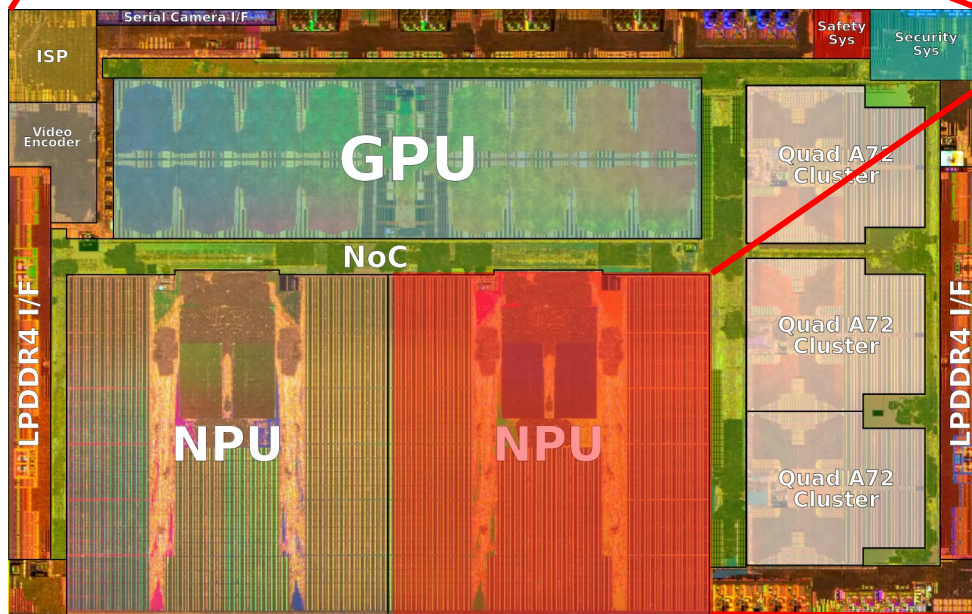


(2020)

Tesla's FSD (Full Self Driving) Computer

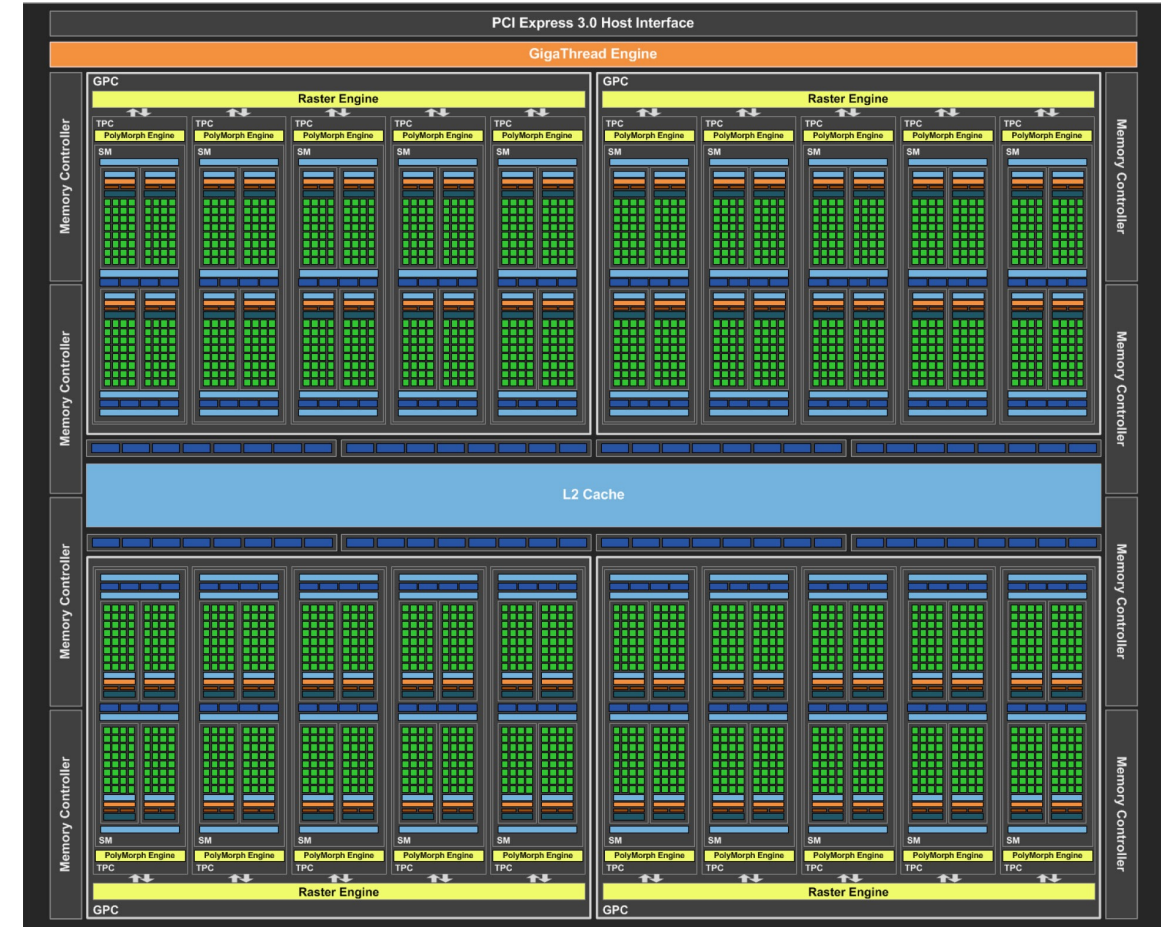
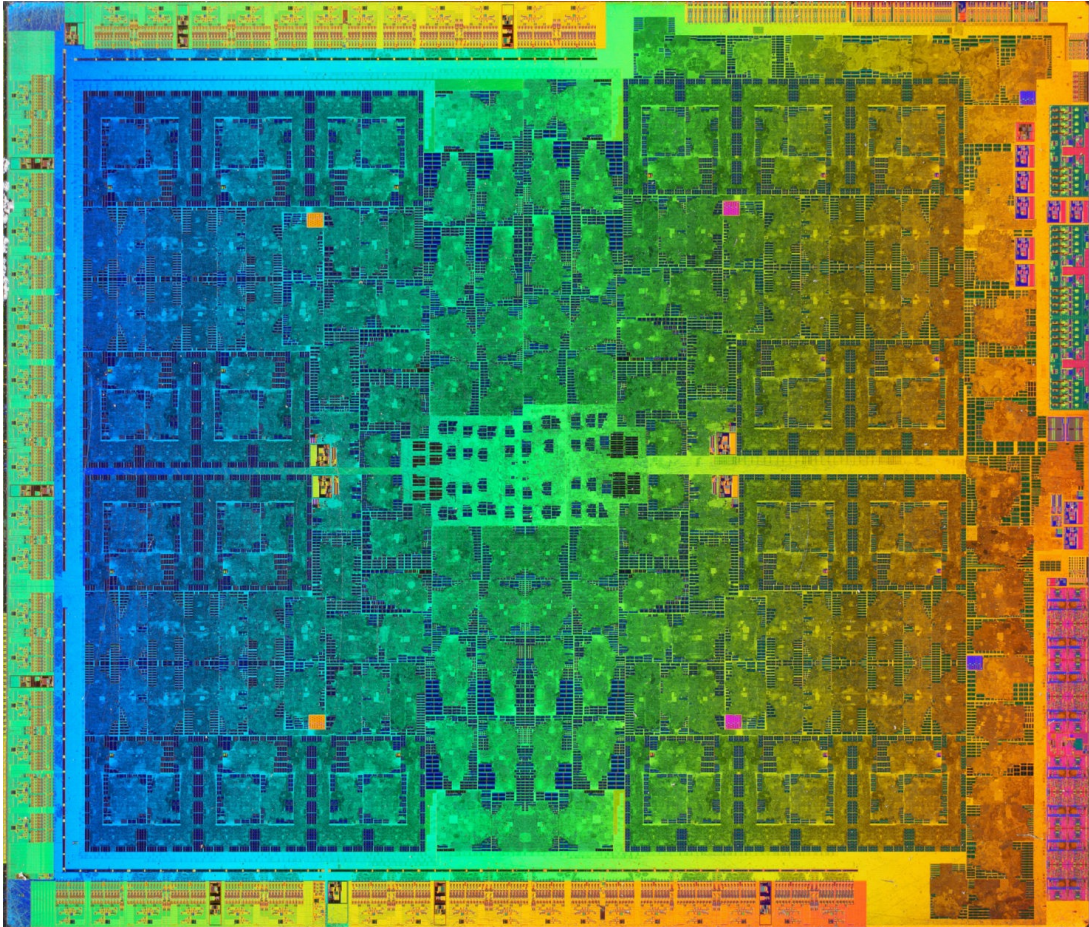


- 147.4 TOPS @ 72 W
- 12 ARM cores x 2
- 1 GPU x 2
- 2 NPU x 2
- Based on Samsung's 14-nm process



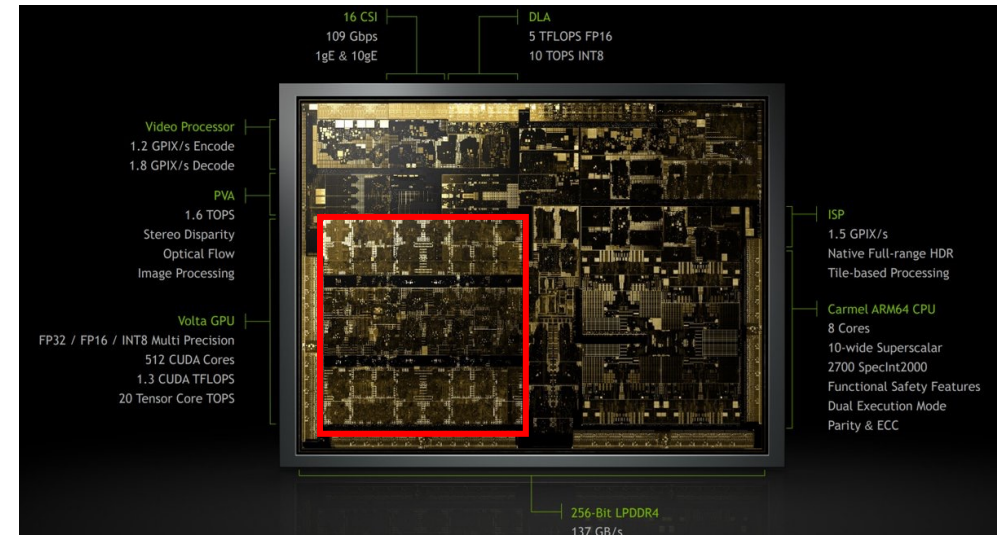
GPU

- Massive number of cores
- Only works as a slave of a (master) CPU

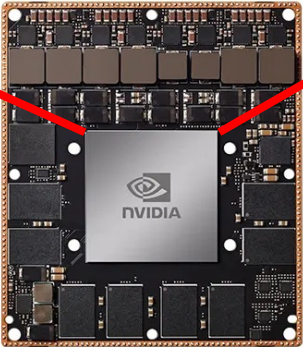
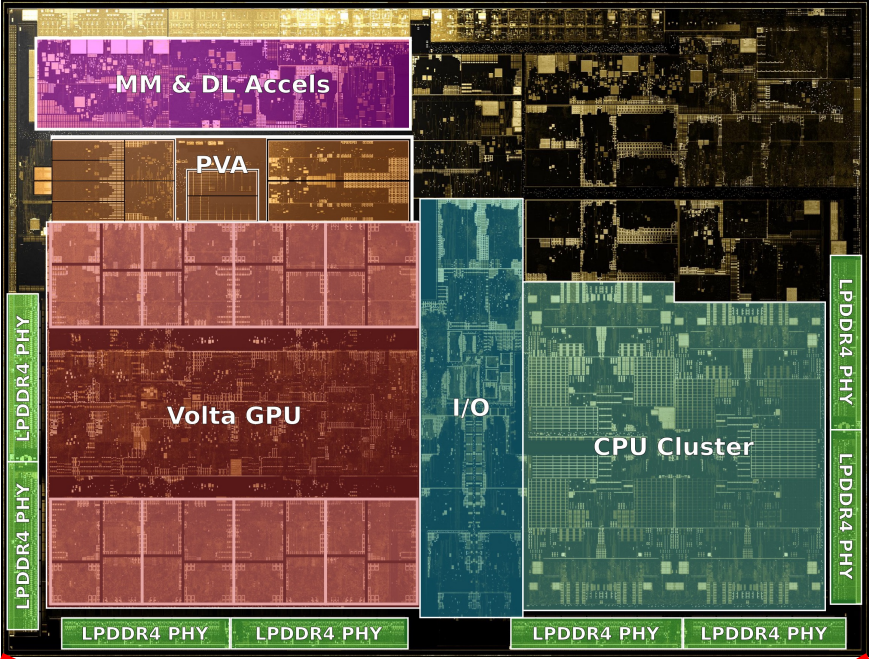
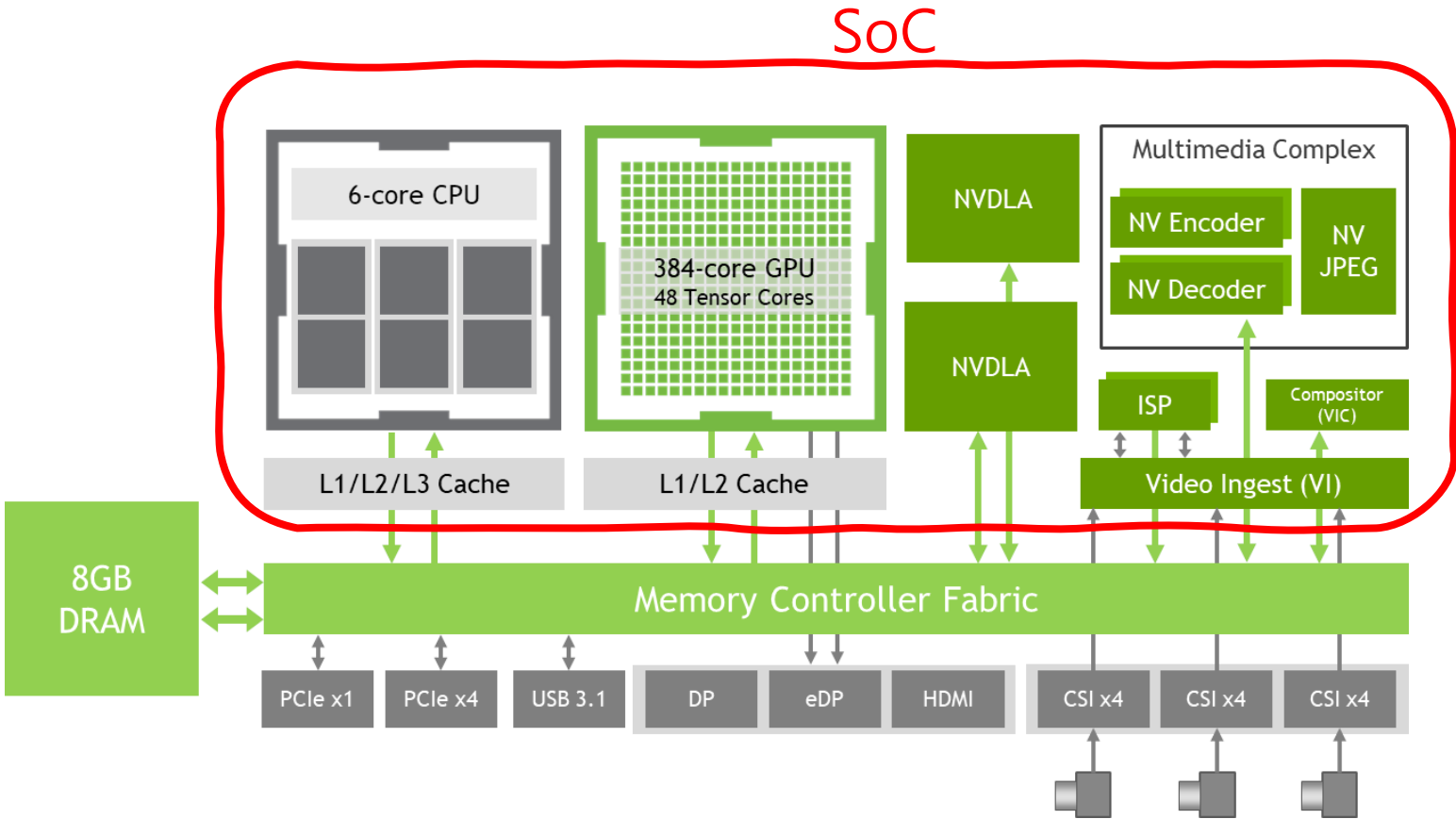


GPU Architectures

- Discrete GPUs (dGPUs)
 - GPUs with dedicated memory
 - PCIe (PCI Express) bus interface
 - Fast but consumes lots of energy
 - For training neural network (NN) models
- Integrated GPUs (iGPUs)
 - GPUs sharing memory with CPU
 - Typically integrated into a SoC die
 - Slow but energy-efficient
 - For inferencing NN models



iGPU Architecture



Jetson AGX Xavier module

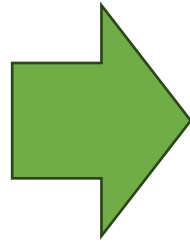
Nvidia Embedded Platforms

1.33 TFLOPs @ 15 W



Jetson TX 2 (2017)

- Quad-core ARM CPU
- 256-core Pascal iGPU

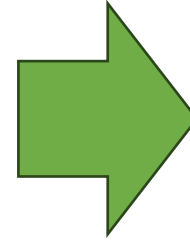


32 TOPS @ 30 W



Jetson AGX Xavier (2018)

- 8-core ARM CPU
- 512-core Volta iGPU



275 TOPS @ 60 W

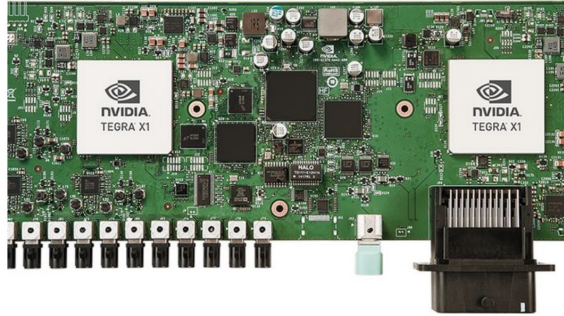


Jetson AGX Orion (2022)

- 12-core ARM CPU
- 2048-core Ampere iGPU

Nvidia Drive Platforms

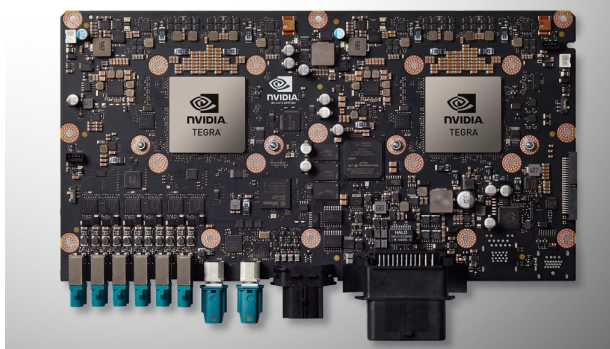
Unknown TOPS @ 20 W



Drive PX (2015)

- 2x Tegra X1 SoC
- 2x Maxwell iGPUs

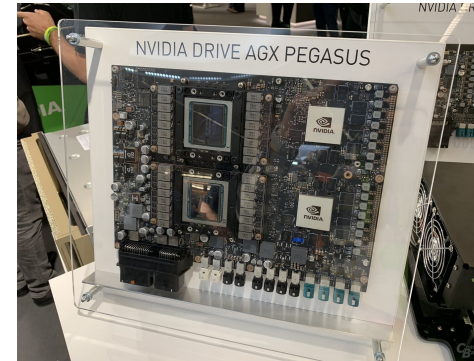
Unknown TOPS @ 80 W



Drive PX 2 (2016)

- 2x Tegra X2 SoC
- 2x Parker iGPUs
- 2x Pascal dGPUs

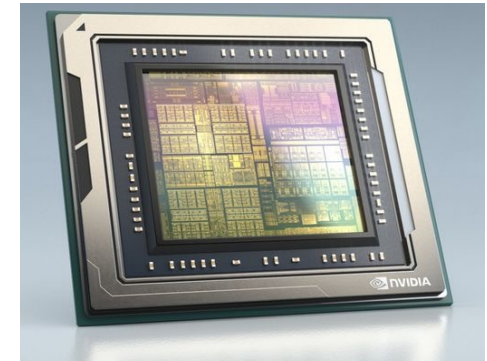
320 TOPS @ 500 W



Drive AGX Pegasus (2017)

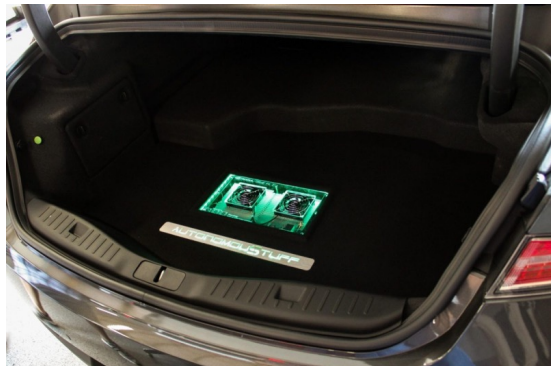
- 2x Tegra Xavier SoC
- 2x Volta iGPUs
- 2x Turing dGPUs

500 TOPS @ 130W



Drive AGX Orin (2022)

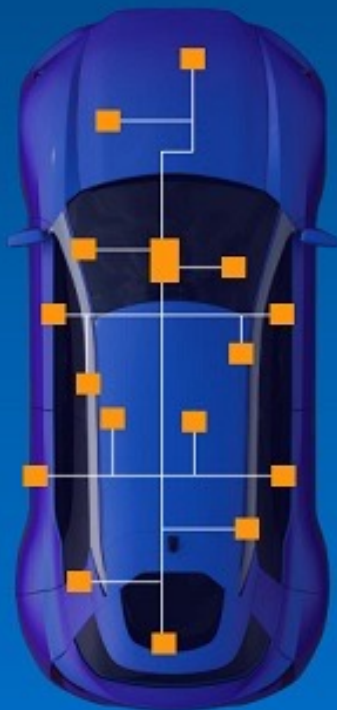
- 2x Tegra Orion SoC
- 2x Ampere iGPUs



Automotive E/E Architecture

CONVERGENCE INTO CAR NETWORK ARCHITECTURE

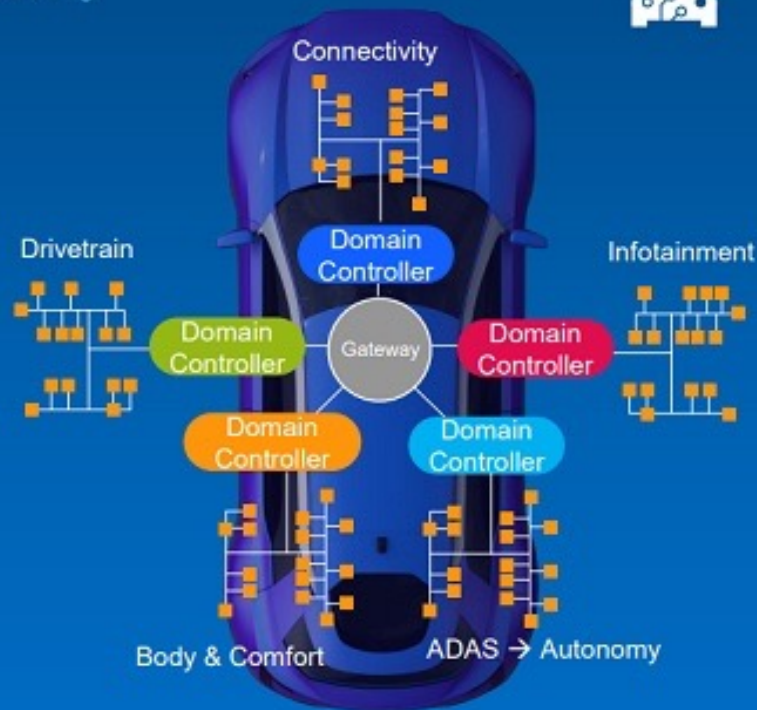
TODAY:
FLAT



Logical: independent functions & SW
Physical: ad-hoc connect [+GW hub]

Logical
restructuring

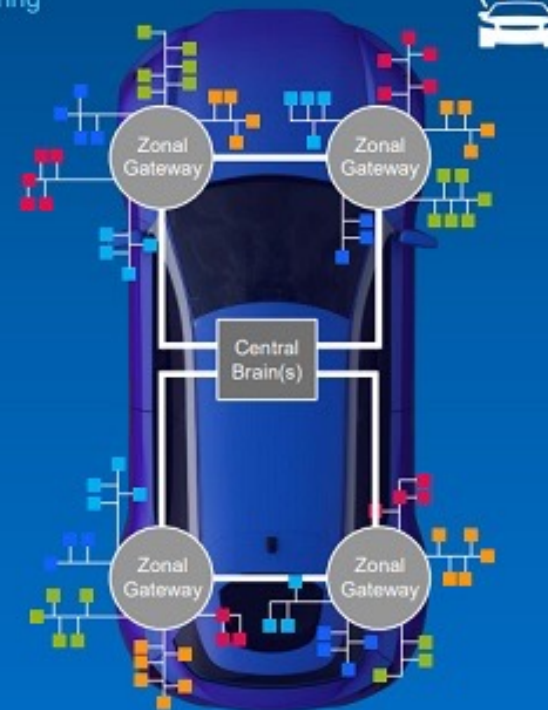
TOMORROW:
DOMAINS



Logical: system hierarchy, specific OSes
Physical: system hard separation

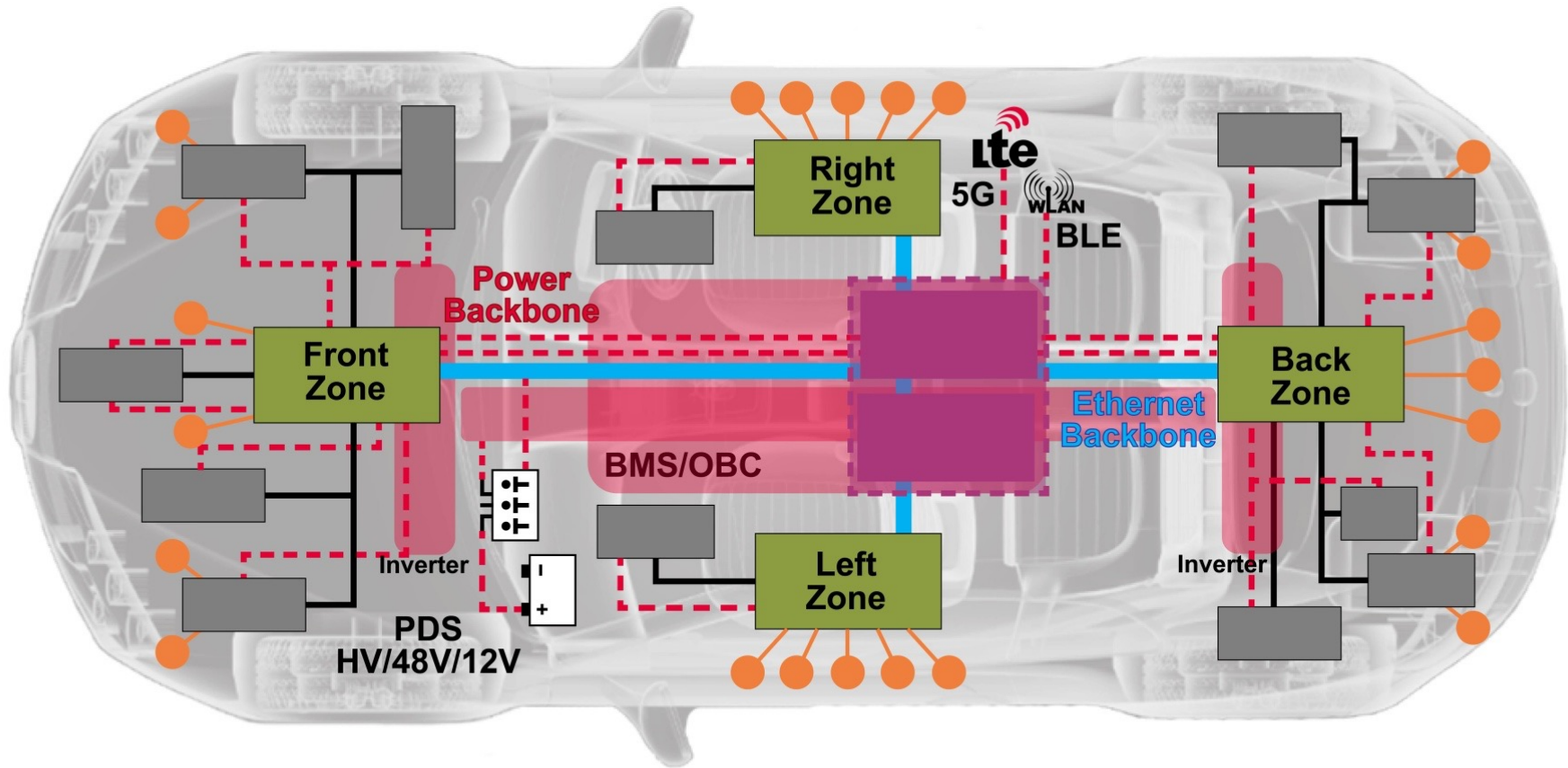
Physical
restructuring

AFTER TOMORROW:
ZONES



Logical: multi-system server(s), one OS
Physical: function-independent rewire

Zonal Architecture



- | | | |
|---|--|---|
|  Automotive Ethernet |  Legacy bus (LIN/CAN..) |  Direct load control or sensor input |
|  Central Computer System |  Zone Controller |  Legacy ECUs |
|  Power Distribution System |  BMS / OBC / Inveter |  Power Supply |

Questions

